SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY - INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Master's Thesis in Informatics

Vision-Language Foundation Models for Multi-Modal Medical Data

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Vision-Language-Foundation-Modelle für Multimodale Medizinische Daten

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I confirm that this master's thesis is my own work and I have documented all sources and material used.

Munich, 16.03.2026 Muhammed Göktepe

Acknowledgments

Abstract

This thesis investigates organ-aware radiology report generation from 3D chest CT using vision-language foundation models. The main challenge is to preserve clinically relevant anatomical information while keeping the visual token interface compact and stable for language decoding.

We propose an end-to-end pipeline that combines: (i) volumetric CT preprocessing and organ-mask supervision, (ii) LLM-based decomposition of reports into organ-level targets, (iii) organ-conditioned visual token extraction from a 3D vision encoder, and (iv) decoder adaptation with parameter-efficient fine-tuning. The framework supports a unified comparison across MedGemma, BERT, GPT-2, and BioBART, with MedGemma-centered ablations completed in this thesis version.

Experiments on CT-RATE show that the strongest gains come from interface design rather than architectural complexity. The best MedGemma variant achieves the top global scores in key metrics, including GREEN (0.348), BLEU-4 (0.105), ROUGE-L (0.379), BERTScore (0.596), RadGraph (0.268), and RadCliQ (1.104), improving over the baseline in most clinically relevant dimensions. Perceiver and multiscale variants did not consistently outperform the simpler baseline under the same training budget, indicating that stable vision-language bridging is a higher-leverage design factor in this setting.

Overall, the thesis provides a reproducible organ-aware 3D report generation framework, code-grounded ablation evidence, and a practical direction for completing cross-decoder comparison with BERT, GPT-2 and BioBART under a shared evaluation protocol.

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1. Introduction

Deep learning has become an important tool across science and engineering, with one of its strongest impacts in computer vision. Today, models perform very well on tasks such as image classification, object detection, and segmentation. These results, however, usually depend on large labeled datasets. In medical imaging, building such datasets is difficult because patient data must be handled carefully and expert annotation is costly and time-consuming.

Within medical imaging, radiology report generation is a high-impact task because reports are the main communication channel between image analysis and clinical decision-making. A useful model must do more than produce fluent text. It must describe findings accurately, preserve anatomical context, and avoid clinically misleading statements. This makes the problem closer to structured clinical reasoning than to generic image captioning.

1.1. Problem Statement

Traditional NLP-based frameworks for radiology report generation often suffered from a lack of localized visual reasoning and a weak integration of clinical knowledge. While early 2D Vision-Language Models (VLMs) leveraging attention mechanisms improved the alignment between image regions and text tokens, they fundamentally failed to capture true three-dimensional anatomical context. In clinical practice, critical findings such as lesion shape, spatial extent, and inter-slice continuity depend entirely on volumetric data.

Applying modern VLM architectures directly to 3D medical volumes, like CT scans, introduces substantial computational bottlenecks. The cubic growth of 3D attention mechanisms requires bridging architectures, such as perceptual resamplers or Q-Formers, to compress thousands of spatial tokens into a dense semantic representation without overwhelming the language decoder. Furthermore, while the field of generative AI has seen an expansion of capable multi-billion parameter foundation models, it remains an open question whether accurate medical report generation depends primarily on the 3D vision encoder or on the semantic reasoning capacity of the Large Language Model decoder.

1.2. Thesis Formulation

This thesis builds on fVLM [1] and focuses on radiology report generation from 3D chest CT. Rather than introducing a new vision backbone, we reuse the pretrained fVLM vision encoder and study how to turn its visual features into clinically reasonable reports.

Our main objective is to improve report quality through design choices around this encoder. Since fVLM does not release report-generation weights and reports limited decoder details, we implement a BERT-base branch [2] with language modeling loss on CT-RATE as a reproducible internal reference aligned with the paper description. We then evaluate additional generation strategies, including organ-conditioned token interfaces and decoder adaptation. Experiments are conducted on the CT-RATE dataset [3], a large-scale public dataset pairing 3D chest CT volumes with free-text radiology reports.

Instead of relying only on text-overlap metrics such as BLEU [4] and ROUGE [5], we also evaluate clinical quality using RadGraph [6] and GREEN [7], the Generative Radiology Report Evaluation and Error Notation metric. This helps us judge whether the generated reports are clinically useful, not only lexically similar.

Through ablations across decoder architectures, including MedGemma [8], BERT-base [2], GPT-2 [9], and BioBART [10], we analyze which decoder choices and interface designs best convert pretrained 3D visual representations into structured clinical text.

1.3. Contributions

The main contributions of this thesis are as follows:

1. **fVLM-Based Report Generation Framework**: We build an end-to-end CT report generation pipeline around a pretrained fVLM vision encoder, with a shared organ-aware visual frontend, decoder-side adaptation, and a reproducible BERT-base reference branch.

2. **Systematic Quality-Improvement Study**: We evaluate practical strategies to improve report generation quality, including organ-conditioned token interfaces, decoder choices such as MedGemma, BERT-base, GPT-2, and BioBART, and training ablations under a controlled setup.

3. **Clinical Efficacy Prioritization**: We validate generated results strictly against clinical-factuality metrics such as GREEN, RadGraph to isolate true clinical reasoning from rote n-gram memorization.

1.4. Thesis Outline

The remainder of this thesis is organized as follows. Chapter 2 reviews relevant literature and motivates the design choices used in this work. Chapter 3 describes the end-to-end pipeline and training setup. Chapter 4 presents the experimental results, Chapter 5 analyzes their implications and limitations, and Chapter 6 summarizes the main outcomes and future directions.

2. Related Work

This chapter reviews the key areas of prior work that form the foundation for this thesis. Section 2.1 discusses the Vision Transformer architecture and its adaptation to medical imaging. Section 2.2 covers vision–language models, including both general-purpose and medical variants. Section 2.3 introduces large language models and parameter-efficient fine-tuning techniques. Section 2.4 surveys radiology report generation methods. Finally, Section 2.5 covers 3D medical image segmentation and organ-guided attention mechanisms.

2.1. Vision Transformers in Medical Imaging

The Vision Transformer (ViT) [11] fundamentally shifted the paradigm of computer vision by applying the transformer architecture, originally designed for natural language processing, to image recognition. ViT divides an input image into fixed-size non-overlapping patches, projects each patch into an embedding, and processes the resulting sequence through a standard transformer encoder. By treating image patches as tokens and relying entirely on multi-head self-attention, ViT eliminates the need for convolutional inductive biases (such as translation invariance and local receptive fields) and achieves competitive or superior performance compared to Convolutional Neural Networks (CNNs) when trained on massive datasets like ImageNet.

2.1.1. Architectural Evolutions: Hierarchical and Self-Supervised ViTs

While the original ViT required large-scale supervised pretraining, subsequent architectures addressed its limitations. The Swin Transformer [12] introduced a hierarchical structure with shifted windows, computing self-attention locally within non-overlapping windows that shift across layers. This hierarchical design reintroduces some useful inductive biases, making it highly efficient for dense prediction tasks like segmentation and object detection, which are critical in medical imaging contexts where precise anatomical boundaries matter.

Simultaneously, advancements in self-supervised learning alleviated the reliance on massive labeled datasets, a severe bottleneck in the medical domain. Masked Autoencoders (MAE) [13] proposed an asymmetric encoder-decoder architecture where the

encoder operates only on an unmasked subset of patches (e.g., 25%), and a lightweight decoder reconstructs the heavily masked (75%) original image. This approach enables the learning of robust, high-level visual representations directly from unlabeled data. Building upon self-supervised paradigms, DINOv2 [14] demonstrated that models trained via self-distillation with no labels can learn universal visual features that are highly transferable across diverse domains, capturing part-level semantic correspondences without explicit supervision.

2.1.2. Adaptation to 3D Medical Volumes

In the medical imaging domain, transformers have been widely adopted to capture the long-range dependencies crucial for understanding complex anatomical structures. Early adaptations like TransUNet [15] combined a CNN encoder for local feature extraction with a transformer bottleneck for global context in 2D medical image segmentation.

However, clinical data such as Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) are inherently 3D volumetric scans. Processing these volume slice-by-slice discards critical through-plane spatial context. UNETR [16] addressed this by using a ViT encoder to process 3D volumetric data directly, extracting 3D patches from the input volume.

The transition from 2D image patches to 3D volumetric patches introduces substantial computational cost. In a standard self-attention mechanism, memory and computational complexity grow quadratically with sequence length. For a 3D medical image, sequence length expands cubically with respect to spatial dimensions. If a high-resolution CT scan is embedded into thousands of 3D tokens, global self-attention can exceed the memory limits of modern GPUs. To mitigate this, modern 3D Vision Transformers commonly use sparse attention, shifted-window variants, or early spatial downsampling.

2.2. Vision–Language Models

Vision–language models (VLMs) aim to bridge the semantic gap between visual perception and natural language understanding. These models learn joint representations of images and text, enabling a wide array of cross-modal tasks such as image captioning, visual question answering (VQA), and complex multi-modal reasoning.

2.2.1. Contrastive Alignment and the CLIP Paradigm

CLIP (Contrastive Language–Image Pre-training) [17] introduced a highly scalable approach to learning visual representations from natural language supervision. By training independent vision and text encoders to maximize the cosine similarity of

matching image–text pairs in a shared embedding space while minimizing the similarity of millions of non-matching pairs, CLIP learns highly transferable visual concepts. Trained on 400 million image–text pairs curated from the internet, CLIP demonstrates strong zero-shot transfer capabilities across a wide range of downstream visual tasks.

Recent advances have refined the CLIP objective. SigLIP (Sigmoid Loss for Language Image Pre-Training) [18] replaces the standard softmax-based contrastive loss with a pairwise sigmoid loss. This change removes the need for global views of all negative pairs across devices, improving training efficiency and scalability and often yielding stronger representations. The contrastive learning paradigm established by CLIP and extended by SigLIP remains a common foundation for aligning vision and language modalities in modern VLMs.

2.2.2. Generative VLMs and Open-Weight Ecosystems

While contrastive models focus purely on alignment and retrieval, generative VLMs extend the paradigm to produce natural language outputs explicitly conditioned on visual inputs. BLIP (Bootstrapping Language-Image Pre-training) [19] proposed a unified framework that jointly trains a contrastive alignment objective, an image–text matching classification objective, and a generative language modeling objective. This multi-task design enables BLIP to be highly effective for both comprehensive image understanding and robust text generation tasks.

Building upon this, BLIP-2 [20] introduced a compute-efficient strategy for bridging the modality gap: the Q-Former (Query Transformer). The Q-Former acts as an informational bottleneck between a frozen image encoder and a frozen large language model. It utilizes a set of learnable query vectors that cross-attend to the dense visual features extracted by the image encoder, forcing the queries to extract only the most text-relevant visual information. By doing so, the Q-Former compresses thousands of spatial visual tokens into a fixed, small number of semantic latent tokens (e.g., 32 tokens). These compressed tokens are then linearly projected directly into the text embedding space of the frozen LLM. This architecture demonstrated that one can achieve strong multi-modal capabilities without unfreezing the massive weights of the underlying language or vision foundation models, reducing computational cost.

A similar design philosophy for handling high-density visual signals was adopted by DeepMind's Flamingo [21]. Rather than a Q-Former, Flamingo utilized a Perceiver Resampler mechanism, which also cross-attends to a large input array (such as a high-resolution image or a varying-length video) using a small set of latent, learnable query vectors. Flamingo then interleaved these compressed visual tokens with standard text tokens inside a frozen LLM through gated cross-attention layers, demonstrating strong few-shot learning and in-context reasoning capabilities.

LLaVA (Large Language-and-Vision Assistant) [22] vastly simplified these complex bridging architectures by proving that a straightforward Multi-Layer Perceptron (MLP) projection layer could effectively map visual features directly from a pretrained CLIP encoder into the native vocabulary embedding space of an LLM, entirely discarding the need for Q-Formers or Resamplers. Through visual instruction tuning (training the model on instruction-following datasets combining images with conversational text), LLaVA achieved remarkable multi-modal dialogue capabilities. Recently, open-source initiatives have rapidly accelerated VLM research. Models like InternVL [23] focus on massive scaling of the vision foundation model to better align with generic linguistic tasks, while the Molmo family [24] emphasizes the crucial role of highly curated, completely open-source point-based multimodal data to achieve state-of-the-art results with highly efficient projector architectures.

2.2.3. Medical Vision-Language Models

Adapting VLMs to the medical domain presents significant unique challenges. Medical images (such as high-resolution multi-parametric MRIs or chest X-rays) have fundamentally different characteristics from natural images (like those in COCO/ImageNet). Furthermore, clinical texts utilize a dense, highly specialized vocabulary, and tolerance for error in medical AI is low.

Early attempts to bridge this gap focused on domain-specific contrastive pretraining. BiomedCLIP [25] addressed this by pretraining a CLIP-style architecture from scratch on 15 million biomedical image–text pairs sourced from PubMed Central articles, demonstrating that in-domain pretraining improves performance on downstream biomedical classification and retrieval tasks compared to generic CLIP models.

For 2D clinical radiology, development has accelerated in recent years [26]. CheXagent [27] was introduced as a clinical foundation model for chest X-ray interpretation, fine-tuned on large corpora of radiology reports to interpret images and answer clinical questions. MAIRA-2 [28] advanced grounded report generation by linking generated assertions to localized bounding boxes in the original X-ray, reducing hallucinations and improving trustworthiness.

For 3D medical imaging, the challenge is compounded by the sheer size of the volumetric data and the scarcity of paired text data. Hamamci et al. [3] introduced CT-CLIP, a contrastive learning framework specifically engineered for 3D computed tomography. Trained exclusively on the massive CT-RATE dataset, which pairs unannotated 3D volumetric CT scans directly with their corresponding free-text radiology reports, CT-CLIP successfully learns 3D structural visual representations that map directly to the clinical text manifold. This foundational work represents one of the first successful large-scale efforts to apply multi-modal vision–language pretraining directly

to unstructured volumetric medical data.

MedGemma [8] is a family of medical foundation models built on top of Google's rigorous Gemma architecture [29]. MedGemma structurally inherits the robust capabilities of the general-purpose Gemma model but undergoes rigorous continued pretraining and specialized instruction-tuning on vast troves of verified biomedical literature, medical board exams, and de-identified clinical notes. The resulting instruction-tuned variant (MedGemma-4B-IT) demonstrates exceptional zero-shot and few-shot capability on complex medical reasoning tasks, making it a highly compelling, medically-aligned language decoder for complex medical VLM pipelines, and ultimately serving as one of the core generative components evaluated in this thesis.

2.3. Decoders and Parameter-Efficient Fine-Tuning

The generation of complex radiology reports from visual features requires a demonstrably powerful language decoder. The development of natural language generation models has largely bifurcated into two fundamentally distinct architectural paradigms: Encoder-Decoder models and Causal Language Models. Understanding the mechanistic differences between these paradigms is crucial when selecting a backbone for medical vision-language reasoning.

2.3.1. Encoder-Decoder Architectures vs. Causal Modeling

Encoder-Decoder models, such as BART (Bidirectional and Auto-Regressive Transformers) [30] and T5, utilize a bidirectional encoder to process the input context comprehensively and an autoregressive decoder to generate the output sequence token by token. Because the encoder can attend to both left and right context simultaneously during the prompt processing phase, these models excel at tasks requiring deep semantic understanding of the entire input condition before generating text (e.g., summarization or machine translation). BART is specifically pretrained by corrupting documents (via token masking, deletion, and sentence permutation) and optimizing a reconstruction loss.

In the biomedical domain, BioBART [10] adapts this exact architecture by continually pretraining the baseline BART model on a massive corpus of PubMed abstracts. This domain-specific pretraining allows BioBART to inherit the robust grammatical generation capabilities of BART while rigidly aligning its vocabulary and internal latent representations with complex, highly specialized medical terminology. In standard vision-language pipelines utilizing Encoder-Decoder backbones, visual embeddings are typically fed directly into the bidirectional encoder, allowing the system to build a rich,

globally aware contextual representation before the autoregressive generation phase even begins.

Conversely, Causal Language Models (CLMs) utilize a decoder-only architecture that predicts the next token in a sequence strictly based on preceding leftward tokens. GPT-2 (Generative Pre-trained Transformer 2) [9] popularized this highly scalable paradigm, demonstrating that simple unsupervised next-token prediction on diverse web corpora could intuitively yield models capable of coherent, zero-shot, open-ended text generation. Due to its relatively modest scale (e.g., 124M parameters for the base model) and exceptionally clean architectural foundation, GPT-2 remains a popular, tractable baseline for experimental diagnostic text generation tasks where compute is constrained.

Unlike bidirectional encoders, causal models process the conditioning prompt (including visual tokens prepended as a "soft prompt") autoregressively. While theoretically limiting the deep bidirectional mingling of context tokens, the massive parameter count of modern causal models overrides this limitation through sheer representational capacity.

Recent foundation models have scaled the decoder-only paradigm to tens of billions of parameters. The LLaMA family [31] and the Gemma family [29] have demonstrated strong capabilities in generalized natural language understanding, logical reasoning, and instruction-following. MedGemma [8], a medical variant (e.g., MedGemma-4B-IT), extends these capabilities with continued medical pretraining and instruction tuning. In causal VLM architectures, preserving the spatial and semantic integrity of the visual token prefix is important because downstream generation depends directly on this initial visual context.

2.3.2. Parameter-Efficient Fine-Tuning (PEFT)

Adapting massive decoders (especially multi-billion parameter foundation models like MedGemma) to specialized downstream tasks such as highly localized, organ-specific radiology report generation via full fine-tuning is computationally prohibitive and can degrade the base model's general linguistic capabilities. Parameter-efficient fine-tuning (PEFT) methods address this by updating only a small fraction of the model's parameters while keeping most pretrained weights frozen.

Early PEFT strategies included Adapter modules [32], which introduced small, trainable feed-forward bottleneck networks sandwiched between the frozen multi-head attention and feed-forward transformer sub-layers. Another approach, Prefix-Tuning [33], mathematically optimizes a sequence of continuous task-specific prompt vectors prepended to the transformer layers. While effective in reducing forgetting, these methods alter the network topology, often introducing additional inference latency

or reducing the available context window depth.

LoRA (Low-Rank Adaptation) [34] addresses these issues and is now widely adopted in foundation-model fine-tuning. LoRA assumes that required updates to dense weight matrices have low intrinsic rank. It introduces trainable low-rank matrices in parallel with frozen linear layers (commonly query and value projections in self-attention). Instead of updating the full matrix $W_0 \in \mathbb{R}^{d \times k}$, LoRA learns smaller matrices $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times k}$ where $r \ll \min(d, k)$, yielding $W_0 + \Delta W = W_0 + BA$. Because BA can be merged into W_0 for deployment, LoRA does not add extra inference latency after merging.

To further reduce training cost, QLoRA [35] combines LoRA with 4-bit Normal Float (NF4) double quantization. In QLoRA, the frozen pretrained model weights reside in 4-bit precision, while only the lightweight, trainable LoRA adapter matrices are cast to higher precision (e.g., BFloat16) during backpropagation. This combination substantially reduces VRAM requirements for fine-tuning multi-billion parameter foundation models, allowing faster iteration on standard hardware.

2.4. Radiology Report Generation

Automatic radiology report generation is the task of producing clinically accurate free-text reports from medical images. This task is fundamentally more challenging than generic image captioning because generated reports must be factually accurate, cover all relevant pathological findings (and explicitly note the absence of others), and strictly adhere to dense clinical conventions.

2.4.1. Traditional and Early Transformer Approaches

Early approaches to radiology report generation adapted encoder–decoder architectures directly from the natural image captioning literature. Jing et al. [36] proposed a co-attention mechanism combined with a hierarchical LSTM decoder to generate structured medical reports from chest X-rays. However, these CNN–RNN frameworks typically produced generic, repetitive descriptions and struggled significantly with capturing rare or subtle findings, often suffering from severe data imbalance in clinical datasets.

Transformer-based methods subsequently improved report generation quality by better tracking long-term context. Chen et al. [37] introduced a memory-driven transformer that used a relational memory module to record key clinical information during autoregressive generation, markedly improving the coherence of long reports. In a follow-up, Chen et al. [38] proposed cross-modal memory networks that maintained separate but interacting memory banks for visual and textual features, enabling more robust cross-modal reasoning. Demonstrating the value of initialization, Nicolson et

al. [39] showed that warm-starting both the vision encoder (e.g., from ImageNet or CheXpert) and the language decoder (e.g., from clinical BERT) substantially improved chest X-ray report generation over training from scratch.

2.4.2. The Evaluation Paradigm: From N-Grams to Clinical Efficacy

Early report-generation studies relied heavily on lexical overlap metrics such as BLEU, ROUGE, and METEOR. In medical reporting, lexical overlap alone is insufficient because clinically meaningful errors can occur with small wording changes. The field therefore shifted toward clinically grounded metrics such as CheXbert [40], RadGraph [6], and GREEN [7], which better reflect factual and relational correctness. In this thesis, the high-level motivation is introduced here, while the operational metric interpretation policy is defined in Section 4.2.1.

2.4.3. 2D Radiology Foundation Models

As the field moved toward the foundation model paradigm, researchers began adapting large, frozen language models as report decoders. R2GenGPT [41] demonstrated that a frozen LLM combined with a trainable visual encoder and a lightweight projection layer could generate high-quality radiology reports with competitive performance on both NLG and clinical efficacy metrics. This paradigm benefits directly from the language understanding capabilities and medical knowledge embedded in pretrained LLMs while keeping multi-modal training costs manageable.

Recently, the field has seen the emergence of true foundation models specifically tailored for 2D X-ray interpretation. CheXagent [27] stands out as a pioneering clinical foundation model designed to interpret chest X-rays. By fine-tuning a massive VLM on diverse corpora of radiology reports and visual question-answering pairs, CheXagent achieved state-of-the-art performance in both free-text report generation and interactive clinical dialogue.

Simultaneously, the architectural focus shifted from mere linguistic fluency to rigorous clinical grounding. MAIRA-2 [28] introduced the concept of grounded radiology report generation. Instead of hallucinating plausible text based on global image context, MAIRA-2 explicitly maps generated clinical assertions in the text back to localized bounding boxes in the original X-ray. This spatial grounding dramatically reduces clinical hallucinations, a critical safety requirement for medical deployment.

2.4.4. 3D Volumetric Report Generation

While 2D X-ray interpretation has matured rapidly, 3D volumetric report generation (e.g., from CT or MRI scans) remains an open and highly challenging frontier. The

jump from 2D to 3D exponentially increases the computational complexity of the visual token sequence. Furthermore, until the recent release of the CT-RATE dataset [3], there was a severe scarcity of paired 3D volumes and free-text reports available for training.

The release of CT-RATE spurred the rapid development of several pioneering 3D multimodal foundation models:

- **CT2Rep** [42]: One of the earliest works to leverage the CT-RATE dataset, CT2Rep demonstrated that automated radiology report generation for 3D medical imaging is feasible by properly projecting 3D visual tokens into a robust language decoder.
- **M3D** [43]: Advancing 3D medical image analysis with Multi-Modal Large Language Models, M3D proposed strategies to efficiently compress volumetric data into spatial tokens, enabling the LLM to process entire CT scans without exceeding context windows.
- **Argus** [44]: Argus introduced comprehensive benchmarking and enhancement strategies for VLMs in 3D radiology, testing models ranging from 3B configurations to massive 70B parameter deployments. Argus highlighted that scaling the LLM capacity yields direct improvements in clinical efficacy metrics like GREEN.
- **Med3DVLM** [45] and **RadFM** [46]: Both models established unified 3D vision-language frameworks, focusing on cross-modal alignment between highly unstructured 3D voxel spaces and precise clinical text.

Recent work has also emphasized fine-grained anatomical supervision for CT understanding. fVLM [1] introduces large-scale and fine-grained vision-language pretraining for CT and reports strong gains on report generation benchmarks. Its organ-aware supervision and decomposition perspective directly motivate the organ-guided decomposition strategy used in this thesis.

A key challenge unifying these works is token compression. Because a raw 3D CT volume contains vastly more information than a 2D image, naive patch extraction results in tens of thousands of tokens that can overwhelm the LLM decoder. Recent studies [47] emphasize that tokenization strategies (such as Perceiver-based latent resampling, spatial 3D pooling, or learned region-of-interest selection) are an important factor for effective 3D VLM performance.

As shown in Table 2.1, under the reported experimental settings, higher resolution inputs and larger language decoders (e.g., Argus-70B) are associated with higher GREEN scores.

Scale note: survey-style tables often report BLEU/GREEN on a percentage scale ($\times 100$), while this thesis reports internal experiment metrics on raw scales in $[0, 1]$.

Table 2.1.: Performance benchmarking of recent 3D Volumetric Foundation Models for Radiology Report Generation evaluated on the CT-RATE dataset. Metrics highlight the paradigm shift toward clinical efficacy (GREEN) over traditional NLP metrics (BLEU-4). Data adapted from recent multi-modal surveys.

Model Name	Parameters	Input Resolution	BLEU-4	ROUGE-L	METEOR	GREEN
Argus-70B	70B	$512 \times 512 \times 256$	47.98	-	-	**60.22**
Argus-8B	8B	$512 \times 512 \times 256$	45.81	-	-	58.42
Argus-3B	3B	$512 \times 512 \times 256$	43.03	-	-	55.56
CT2Rep-3B	3B	$256 \times 256 \times 64$	38.77	48.96	-	51.72
M3D-3B	3B	$256 \times 256 \times 64$	40.23	50.31	-	50.31
RadFM-3B	3B	$256 \times 256 \times 64$	39.95	49.46	-	49.46
COLIPRI-CRM	1B	$160 \times 160 \times 160$	45.20	55.20	57.80	-

2.5. 3D Medical Image Segmentation and Organ-Guided Attention

Accurate anatomical localization is an essential prerequisite for generating highly specific, clinically actionable radiology reports. The availability of robust, automated whole-body 3D segmentation tools has enabled novel VLM architectures that explicitly leverage anatomical priors to ground their text generation.

2.5.1. Volumetric Segmentation Frameworks

While early segmentation models relied on 3D CNNs (e.g., 3D U-Net), transformer-based architectures have increasingly dominated this space by effectively capturing long-range contextual dependencies. TransUNet [15] and Swin-UNet [48] demonstrated the efficacy of transformers for 2D medical segmentation, but volumetric data requires dedicated 3D processing.

nnFormer (Interleaved Transformer for Volumetric Medical Image Segmentation) [49] extends transformer capabilities explicitly to 3D medical images. By interleaving local 3D convolution operations with global multi-head self-attention mechanisms, nnFormer captures both the intricate local textures of tumors and the global geometrical relationships of large organs, setting strong benchmarks across multiple 3D medical datasets.

In practical clinical pipelines, TotalSegmentator [50] has emerged as an indispensable, widely used utility for automatic segmentation. Built upon the robust nnU-Net framework, TotalSegmentator provides highly reliable, out-of-the-box segmentation of 104 distinct anatomical structures, including major organs, bones, vessels, and muscles,

from standard CT images. The discrete 3D masks produced by such algorithms can serve directly as precise anatomical priors for downstream generative tasks.

2.5.2. Organ-Guided Attention and Clinical Grounding

Incorporating these anatomical priors into VLM architectures enables a shift from global image captioning to localized, organ-specific reasoning. Rather than processing an entire CT volume as a uniform sequence of visual tokens, organ-guided mechanisms restrict or bias the model's cross-attention to regions corresponding to specific query structures (e.g., the lungs or the liver).

This approach draws heavily upon learnable query architectures such as the Perceiver [51], where a fixed set of latent vectors attends to a massive sensory input. In the medical domain, these latent vectors can represent specific organs. By masking the visual features using 3D segmentation maps prior to cross-attention, the language decoder is forced to focus only on the relevant anatomy when generating the corresponding sentence in the radiology report.

Furthermore, in the context of multi-task learning [52], organ-specific feature extraction can be significantly enhanced via auxiliary classification objectives. By jointly training the visual encoder to perform specific disease classification tasks alongside open-ended report generation, the model learns a more disentangled and discriminative organ-level representation. This auxiliary grounding acts as a powerful inductive bias, encouraging the visual encoder to capture the subtle physiological abnormalities that drive accurate clinical reporting.

3. Methodology

3.1. Pipeline Overview

This thesis studies organ-aware report generation from 3D chest CT. The model input is a CT volume $V \in \mathbb{R}^{D \times H \times W}$ together with organ masks $\{M_o\}_{o \in \mathcal{O}}$ for a fixed organ set $\mathcal{O}$. The output is an organ-structured report in which each organ receives a dedicated text segment.

For patient i, we model:

$$\hat{Y}_i = f_\theta(V_i, \{M_{i,o}\}_{o \in \mathcal{O}}), \tag{3.1}$$

where $\hat{Y}_i = \{\hat{y}_{i,o}\}_{o \in \mathcal{O}}$ denotes organ-level generated text. A report-level prediction is assembled by concatenating organ sections in a fixed anatomical order.

The end-to-end workflow contains six stages:

1. CT preprocessing and mask preparation.
2. Report decomposition into organ-level supervision.
3. Organ-aware visual token extraction from a 3D ViT encoder.
4. Projection/bridging into decoder embedding space.
5. Organ-conditioned text generation.
6. Report assembly and metric computation.

The same interface is used across four decoder families: MedGemma [8], BERT-base [2], GPT-2 [9], and BioBART [10].

3.2. Data and Targets

3.2.1. Volume and Mask Preprocessing

The training and evaluation code uses a consistent MONAI-based transform stack. Given raw NIfTI CT and segmentation volumes, preprocessing applies:

1. channel standardization and axis transposition,

2. HU clipping ($[-1150, 350]$) and min-max normalization to $[0, 1]$,
3. spatial harmonization to $(112, 256, 352)$ via padding/resizing,
4. trilinear interpolation for CT and nearest-neighbor interpolation for masks.

Replacing center-cropping with full-volume resizing reduces accidental exclusion of peripheral anatomy.

3.2.2. Target Organ Set and Segmentation Mapping

The final schema uses 12 organs:

$$\mathcal{O} = \{\text{lung, heart, esophagus, liver, gallbladder, stomach, pancreas, spleen, kidney, aorta, trachea, rib}\}. \tag{3.2}$$

Mask extraction follows fixed segmentation-ID mappings summarized in Table 3.1.

Table 3.1.: Organ-to-segmentation-ID mapping used for binary mask construction.

Organ	**Segmentation IDs**
Lung	10, 11, 12, 13, 14
Heart	51, 61
Aorta	52
Esophagus	15
Trachea	16
Rib	92–115
Liver	5
Gallbladder	4
Stomach	6
Pancreas	7
Spleen	1
Kidney	2, 3

3.2.3. Report Decomposition Targets

Reference reports are decomposed into organ-keyed JSON records using an instruction-following LLM pipeline (Llama-3.3-70B-Instruct via vLLM [53], with AWQ quantization [54]). The decomposition enforces:

1. schema-constrained JSON output,

2. one text field per organ key,
3. explicit empty/null handling for missing organs.

This converts free-text supervision into organ-conditioned targets and enables organ-level error analysis.

During training, each organ is paired with an organ-specific prompt. MedGemma branches use chat-format prompts; BERT-base/GPT-2/BioBART branches use shorter textual prompts (e.g., `Describe [organ]:`). In all branches, loss is masked to the response region only.

3.2.4. Chest-CT Visibility Caveat

Although the dataset modality is chest CT, several upper-abdominal organs appear in a non-trivial subset of scans. Visibility remains inconsistent across patients and protocols, so sparse organs are treated as optional supervision targets rather than mandatory output constraints.

3.3. Model Architecture

3.3.1. 3D Visual Backbone

All principal variants use a 3D ViT encoder initialized from pretrained weights and fine-tuned end-to-end. The standard configuration uses input size $(112, 256, 352)$, patch size $(16, 16, 32)$, and hidden size 768.

3.3.2. Organ Queries and Masked Cross-Attention

Let $F(V) \in \mathbb{R}^{T \times d_v}$ be ViT token features and $Q_o \in \mathbb{R}^{K \times d_v}$ be learned queries for organ o. Organ masks are downsampled to the ViT token grid and converted into additive attention bias B_o that blocks background tokens.

Organ token extraction is:

$$Z_o = \mathrm{MHA}(Q_o, F(V), F(V); B_o), \tag{3.3}$$

where $Z_o \in \mathbb{R}^{K \times d_v}$. In implementation, mask downsampling is aligned to the token-grid geometry $(7, 16, 11)$.

3.3.3. Token Granularity and Empty-Organ Safety

Two token-interface regimes are used:

- **Multi-token interface:** $K = 8$ tokens per organ (96 tokens for 12 organs).
- **Compressed interface:** effective one-token-per-organ representation before decoder conditioning.

Empty masks are explicitly handled. If an organ query sees only blocked positions, attention bias is reset for that path to avoid undefined softmax behavior and NaN propagation.

3.3.4. Projection, Stabilization, and Sentinel Wrapping

Visual features are projected from $d_v = 768$ to decoder hidden size d_l:

$$\tilde{Z}_o = W_p Z_o + b_p, \quad W_p \in \mathbb{R}^{d_v \times d_l}. \tag{3.4}$$

Before concatenation with text embeddings, two stabilization steps are applied:

1. layer normalization on projected visual tokens,
2. variance matching to decoder embedding scale.

For MedGemma, the sequence format is:

$$[\text{BOS}],\ \texttt{<vis>},\ \tilde{Z}_o,\ \texttt{<end_vis>},\ \text{text prompt and target tokens.} \tag{3.5}$$

Loss is masked over injected visual tokens so gradients are concentrated on target text generation.

3.4. Variants Evaluated

Table 3.2 summarizes the major variants. The baseline (`medical_vlm_8_tokens_full`) uses masked organ cross-attention and a direct projector. The strongest MedGemma variant (`medgemma_lora_vis_token_pos_embed`) adds visual positional embeddings and explicit visual sentinels. Perceiver and multiscale variants test heavier visual aggregation/fusion. The v3 branch adds a CNN stem and alignment regularization. Curriculum and hard-example branches isolate optimization-policy effects under the same core interface. Decoder-family branches cover MedGemma, BERT-base, GPT-2, and BioBART with the same visual frontend.

Table 3.2.: Model variants and principal design differences.

Variant	Main Design Choice
`medical_vlm_8_tokens_fine`	Baseline MedGemma branch with 8 organ tokens, masked cross-attention, projector, and weighted LM training.
`medgemma_lora_vis_tokens_pos_emb`	Adds visual positional embeddings and explicit visual sentinel wrapping; strongest global performance in current runs.
`perceiver_resampler`	Replaces single cross-attention with stacked Perceiver-style self/cross-attention blocks.
`multiscale_vit_fpn`	Fuses intermediate ViT features from multiple depths before organ-query attention.
`medgemma_architecture_v3`	Adds CNN stem and visual-text alignment loss; explores stronger encoder bias and contrastive regularization.
`curriculum_learning`	Two-phase pathology-first curriculum before full distribution training.
`hard_example_mining`	Dynamic weighted sampler emphasizing high-loss samples.
BERT-base / GPT-2 / BioBART branches	Same organ-attention encoder interface with decoder-family adaptation for cross-family comparison.

3.5. Decoder Adaptation and Training Objective

3.5.1. Decoder-Specific Adaptation

MedGemma is adapted with LoRA [34] while the base model remains frozen. Main settings are rank $r = 16$, $\alpha = 32$, dropout 0.05, and target modules `q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj`.

BERT-base, GPT-2, and BioBART branches use the same organ-attention visual frontend within a `VisionEncoderDecoderModel` interface. The BERT-base branch is trained with language modeling loss as a reproducible reference aligned with the fVLM decoder description. Decoder-side adaptation unfreezes a restricted subset (cross-attention/normalization/output layers and embeddings). For BioBART, shared embeddings and output-head weights are copied from pretrained checkpoints to avoid random-initialization drift in wrapper mode.

3.5.2. Weighted Language Modeling Loss

With batch index i, organ o, and token position t, the primary objective is:

$$\mathcal{L}_{\text{LM}} = \frac{1}{B|\mathcal{O}|} \sum_{i=1}^{B} \sum_{o \in \mathcal{O}} w_{i,o} \left(\frac{1}{T_{i,o}} \sum_{t=1}^{T_{i,o}} \text{CE}(y_{i,o,t}, \hat{y}_{i,o,t}) \right), \tag{3.6}$$

where $w_{i,o}$ is an organ/sample weight for imbalance control.

3.5.3. Optional Alignment Objective

Some variants add visual-text contrastive alignment:

$$S_{o,o'} = \frac{\langle v_o, t_{o'} \rangle}{\tau}, \tag{3.7}$$

and optimize:

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda \mathcal{L}_{\text{align}}. \tag{3.8}$$

This is used mainly in `medgemma_architecture_v3` and selected BERT-base/GPT-2/BioBART branches.

3.6. Optimization and Inference

3.6.1. Shared Optimization Settings

Across main retraining runs, settings are largely standardized:

- batch size 1 per device, gradient accumulation 8,
- learning rate 1×10^{-4}, weight decay 0.01, warmup ratio 0.05,
- 2 retraining epochs,
- bf16 with gradient checkpointing,
- checkpoint/evaluation every 200 steps.

3.6.2. Training-Strategy Variants

Beyond architecture, the thesis evaluates curriculum learning, hard-example mining, undersampling of dominant "no-finding" targets, and organ-weighted losses.

3.6.3. Organ-Wise Inference and Report Assembly

At inference, one prompt is generated per organ and decoded with beam search. MedGemma runs typically use 3 beams and `max_new_tokens=100`; BERT-base/GPT-2/BioBART runs typically use 4 beams and `max_length=120`. Outputs are assembled in fixed organ order:

$$\hat{R} = \texttt{LUNG:}\ \hat{y}_{\text{lung}} \parallel \cdots \parallel \texttt{KIDNEY:}\ \hat{y}_{\text{kidney}}. \tag{3.9}$$

Organs with weak or absent visual support are not forced to produce abnormal statements, reducing schema-induced hallucination.

3.7. Reproducibility Notes

- Training/evaluation artifacts are isolated per variant to prevent checkpoint and metric contamination.
- Official runs include generated report files and standardized `metrics_final.csv` summaries.
- Metric computation is centralized through one RadEval-based script with fixed key mapping.
- Decoder comparison uses only official organ-attention result folders.

This design keeps the thesis contribution focused: controlling organ grounding and token interface under a consistent data protocol, then comparing architecture and decoder choices on that shared foundation.

4. Experiments and Results

4.1. Experimental Setup

4.1.1. Objectives and Research Questions

This chapter evaluates organ-aware 3D CT report generation under a consistent CT-RATE protocol. Results are organized by four research questions:

- **RQ1 (Token Interface):** Does organ-token interface design improve report quality?
- **RQ2 (Architecture):** Do heavier visual modules outperform the simpler masked cross-attention baseline?
- **RQ3 (Training Strategy):** How much do curriculum/hard-example/imbalance policies change outcomes?
- **RQ4 (Decoder Family):** Under the same frontend, how do MedGemma, GPT-2, and BioBART compare?

4.1.2. Dataset and Preprocessing

All experiments use **CT-RATE** [3], with official split conventions [55]. Table 4.1 summarizes dataset-level context.

Table 4.1.: CT-RATE dataset-level statistics used in this thesis.

Statistic	Value
Total CT Volumes	25,692
Unique Patients	21,304
Modality	3D chest CT
Text Modality	Findings + impression
Split Policy	Official train/validation/test split

Preprocessing follows Chapter 3: HU clipping/normalization, spatial harmonization to $(112, 256, 352)$, organ-mask extraction, and report decomposition into organ sections.

4.1.3. Organ Coverage Profile

Per-organ support is uneven in decomposed chest-CT reporting. Table 4.2 reports matched-reference counts from the best MedGemma evaluation artifact (same pipeline used for all metrics).

Table 4.2.: Per-organ coverage among matched validation samples (GLOBAL=1563).

Organ	N	Coverage (%)
Lung	1562	99.94
Heart	1446	92.51
Esophagus	1353	86.56
Trachea	1411	90.28
Aorta	867	55.47
Liver	850	54.38
Kidney	280	17.91
Gallbladder	153	9.79
Spleen	146	9.34
Rib	117	7.49
Pancreas	57	3.65
Stomach	20	1.28

4.1.4. Implementation and Optimization Defaults

Training/evaluation uses PyTorch, HuggingFace Transformers, and MONAI. Default optimization settings across retraining runs are listed in Table 4.3.

4.1.5. Ablation Axes

Table 4.4 defines the comparison axes used throughout this chapter.

4.2. Evaluation Protocol

4.2.1. Metric Families

We report complementary metric groups: lexical overlap (BLEU-4 [4], ROUGE-L [5], METEOR [56], CIDEr [57]), semantic similarity (BERTScore [58]), and clinical/factual consistency (CheXbert [40], RadGraph [6], GREEN [7], RadCliQ [59], SRR-BERT style score).

Table 4.3.: Shared optimization defaults across main retraining runs.

Hyperparameter	**Value**
Per-device batch size	1
Gradient accumulation	8
Learning rate	1×10^{-4}
Weight decay	0.01
Warmup ratio	0.05
Epochs (retrain stage)	2
Precision	bf16
Gradient checkpointing	Enabled
Save/eval frequency	Every 200 steps

Table 4.4.: Ablation axes and evaluated variants.

Axis	**Variants**
Token/bridge design	`medical_vlm_8_tokens_full`, `medgemma_lora_vis_token_pos_embed`, `medical_vlm_8_tokens_full_maxpool`
Encoder architecture	`perceiver_resampler`, `multiscale_vit_fpn`, `medgemma_architecture_v3`
Training strategy	`curriculum_learning`, `hard_example_mining`, `lora_with_vis_tokens_pos_embed_undersampling`, `lora_with_vis_tokens_pos_embed_weight_loss`
Decoder family	MedGemma, GPT-2, BioBART under shared organ-attention frontend
External contextual baseline	fVLM [1] (different protocol; contextual only)

Formal equations are provided in Appendix A. Chapter-level interpretation focuses on metric behavior and agreement patterns.

4.2.2. Metric Interpretation Policy

Model ranking is anchored on GREEN and RadGraph, then cross-checked with lexical-semantic metrics. CheXbert is reported but interpreted cautiously for organ-decomposed CT text because its design assumptions are closer to full-report label extraction workflows.

4.2.3. Evaluation Pipeline and Artifact Standardization

For each model checkpoint:

1. Generate reports and export `generated_reports.csv` (or `generated_reports_gemma.csv`).
2. Compute metrics with unified `radeval_metrics.py`.
3. Export standardized `metrics_final.csv` with GLOBAL and per-organ rows.

The key mapping from RadEval output to thesis columns is fixed (e.g., `green` → GREEN, `radgraph_simple` → RadGraph, `radcliq-v1` → RadCliQ), ensuring direct comparability across variants.

4.2.4. Source-of-Truth Policy

- MedGemma ablations: `fvlm/rep_medgemma/results_retrain/global_model_comparison.csv`.
- Decoder comparison: official `rep_vision_organ_attention/results/gpt2_v3` and `.../biobart_v3` artifacts.
- External baseline: fVLM paper statistics [1] in a separate contextual block.

4.2.5. Qualitative Protocol for Attention Maps

Cross-attention maps are interpreted with the following checks: spatial overlap with the queried organ mask, compactness vs diffusion, stability across planes, and false activation in anatomically confounding neighbors.

4.3. Results by Research Question

4.3.1. Main MedGemma Result Tables

Table 4.5 reports the core MedGemma ablations, and Table 4.6 provides the full aggregated matrix.

Table 4.5.: Core MedGemma ablations on CT-RATE validation set (GLOBAL metrics).

Model	GREEN	BLEU-4	METEOR	ROUGE-L	CIDEr	BERTScore	RadGraph	CheXbert	RadCliQ	SRR-BERT
medical_vlm_8_tokens_full (baseline)	0.323	0.092	0.328	0.341	0.041	0.567	0.243	0.239	1.037	**0.330**
medgemma_lora_vis_token_pos_embed (best)	**0.348**	**0.105**	**0.344**	**0.379**	**0.084**	**0.596**	**0.268**	0.026	**1.104**	0.324
multiscale_vit_fpn	0.327	0.086	0.322	0.337	0.043	0.566	0.242	0.241	1.022	0.326
perceiver_resampler	0.316	0.082	0.317	0.333	0.036	0.564	0.236	0.221	1.008	0.308
medgemma_architecture_v3	0.313	0.083	0.318	0.335	0.030	0.564	0.242	0.206	1.007	0.301
curriculum_learning	0.194	0.032	0.250	0.250	0.018	0.507	0.133	**0.269**	0.773	0.322

Table 4.6.: Full MedGemma retraining matrix (GLOBAL rows from official aggregate file).

Model	GREEN	BLEU-4	METEOR	ROUGE-L	CIDEr	BERTScore	RadGraph	CheXbert	RadCliQ	SRR-BERT
curriculum_learning	0.194	0.032	0.250	0.250	0.018	0.507	0.133	0.269	0.773	0.322
hard_example_mining	0.316	0.083	0.319	0.332	0.043	0.563	0.237	0.198	0.999	0.307
lora_with_vis_tokens_pos_embed_undersampling	0.318	0.082	0.316	0.334	0.028	0.563	0.237	0.203	1.000	0.297
lora_with_vis_tokens_pos_embed_weight_loss	0.298	0.063	0.294	0.316	0.025	0.555	0.214	0.199	0.952	0.293
medgemma_alignment_v1	0.313	0.083	0.318	0.332	0.032	0.564	0.238	0.216	1.007	0.316
medgemma_architecture_v3	0.313	0.083	0.318	0.335	0.030	0.564	0.242	0.206	1.007	0.301
medgemma_lora_vis_token_pos_embed	**0.348**	**0.105**	**0.344**	**0.379**	**0.084**	**0.596**	**0.268**	0.026	**1.104**	0.324
medical_vlm_8_tokens_full	0.323	0.092	0.328	0.341	0.041	0.567	0.243	0.239	1.037	**0.330**
medical_vlm_8_tokens_full_maxpool	0.320	0.086	0.322	0.338	0.035	0.566	0.241	0.213	1.020	0.316
multiscale_vit_fpn	0.327	0.086	0.322	0.337	0.043	0.566	0.242	**0.241**	1.022	0.326
perceiver_resampler	0.316	0.082	0.317	0.333	0.036	0.564	0.236	0.221	1.008	0.308

4.3.2. RQ1: Token Interface Design

RQ1 is answered positively. The positional/sentinel visual-prefix variant (`medgemma_lora_vis_token_pos_` is the strongest global model and improves GREEN, RadGraph, and most lexical-semantic metrics over the baseline.

4.3.3. RQ2: Architecture Complexity

Table 4.7 summarizes deltas vs baseline for representative models. Heavier modules (Perceiver, multiscale, v3) do not consistently outperform the simpler interface-first baseline under the current training budget.

4.3.4. RQ3: Training Strategy

Training-policy interventions (hard-example mining, undersampling, weighted-loss variants) are competitive but do not surpass the best interface variant. Curriculum

Table 4.7.: Delta against baseline (GLOBAL metrics). Positive is better.

Model	ΔGREEN	ΔBLEU-4	ΔRadGraph	ΔRadCliQ
medgemma_lora_vis_token_pos_embed	+0.0249	+0.0131	+0.0250	+0.0668
multiscale_vit_fpn	+0.0045	-0.0065	-0.0015	-0.0156
perceiver_resampler	-0.0070	-0.0099	-0.0068	-0.0298
medgemma_architecture_v3	-0.0097	-0.0098	-0.0009	-0.0302
hard_example_mining	-0.0071	-0.0088	-0.0064	-0.0386
curriculum_learning	-0.1286	-0.0602	-0.1100	-0.2647

learning degrades most global metrics in the current two-phase configuration.

4.3.5. RQ4: Decoder Family Comparison

Decoder comparison is a core axis with fixed protocol and table schema. Current row status is shown in Table 4.8. As of March 3, 2026, official GPT-2/BioBART folders contain generated reports but not finalized `metrics_final.csv`; rows will be populated immediately after standardized metric extraction.

Table 4.8.: Decoder comparison on CT-RATE under shared organ-attention frontend.

Decoder	**GREEN**	**BLEU-4**	**METEOR**	**ROUGE-L**	**CIDEr**	**BERTScore**	**RadGraph**	**CheXbert**	**RadCliQ**	**SRR-BERT**
MedGemma (best variant)	0.348	0.105	0.344	0.379	0.084	0.596	0.268	0.026	1.104	0.324
GPT-2 (`results/gpt2_v3`)	–	–	–	–	–	–	–	–	–	–
BioBART (`results/biobart_v3`)	–	–	–	–	–	–	–	–	–	–

To avoid mixing unofficial outputs, only structural diagnostics are reported before final metric pass (Table 4.9).

Table 4.9.: Decoder run-status diagnostics from official generated report files (not final quality metrics).

Decoder	**Samples**	**Avg words/report**	**Avg organ headers**	**12-header completeness (%)**
MedGemma best	1564	157.9	12.00	100.00
GPT-2 v3	1563	435.0	5.29	0.38
BioBART v3	1563	357.8	5.29	0.38

4.4. Fine-Grained Analyses

4.4.1. Organ-Level Analysis

Table 4.10 reports selected organ-level metrics for the best MedGemma variant.

Table 4.10.: Best MedGemma variant: selected organ-level metrics.

Organ	N	GREEN	RadGraph	CheXbert
LUNG	1562	0.169	0.074	0.060
HEART	1446	0.626	0.502	0.000
ESOPHAGUS	1353	0.651	0.426	0.831
TRACHEA	1411	0.817	0.318	0.957
AORTA	867	0.351	0.125	0.636
LIVER	850	0.561	0.593	0.690
KIDNEY	280	0.134	0.220	0.403

Thoracic organs with strong support tend to show higher stability than low-support abdominal organs.

4.4.2. Cross-Attention Activation Case Studies

Cross-attention overlays are extracted with `fvlm/rep_medgemma/visualize_attention.py`. They are used to assess localization quality, robustness under multi-region findings, and behavior differences across model variants.

Case Study A: Typical Thoracic Pattern

Case Study B: Mixed Thoracic and Upper-Abdominal Findings

Case Study C: Baseline vs Perceiver vs Multiscale

4.4.3. External Contextual Baseline: fVLM

Table 4.11 reproduces the report-generation results from fVLM Table 3 [1] (MedVL-CT69K). These values are reported on a percentage scale (0–100). They are included as an external contextual baseline, not as a direct row-wise comparator to our CT-RATE results.

In the fVLM source table, the strongest overall row is *Fine-tuning / Ours*, with the only metric exception that Merlin is marginally higher on CIDEr (17.2 vs 17.1). Under

Placeholder
Case A – Baseline attention map
(medical_vlm_8_tokens_full)

(a) Baseline model

Placeholder
Case A – Best model attention map
(medgemma_lora_vis_token_pos_embed)

(b) Best MedGemma variant

Figure 4.1.: Case Study A: baseline vs best model cross-attention overlays in a thoracic-dominant exam.

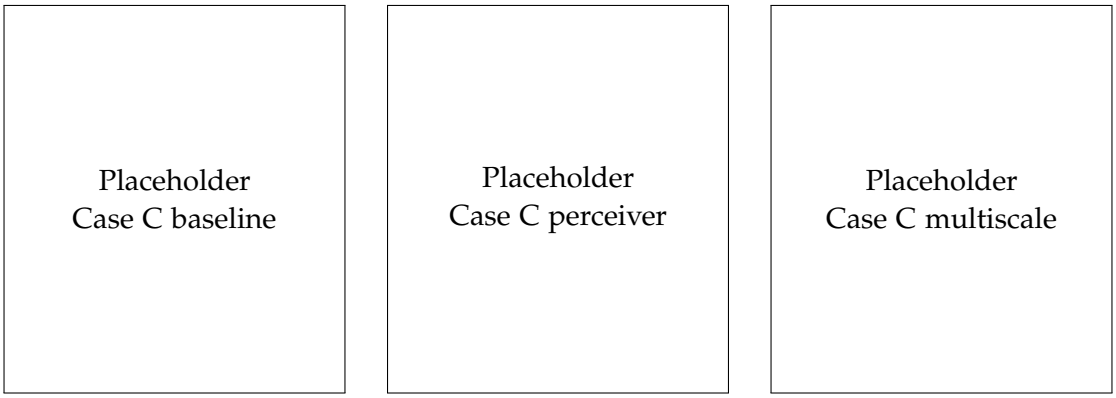
Placeholder
Case B – Baseline attention map
(thorax + upper abdomen visible)

(a) Baseline model

Placeholder
Case B – Best model attention map
(thorax + upper abdomen visible)

(b) Best MedGemma variant

Figure 4.2.: Case Study B: behavior under mixed thoracic and upper-abdominal findings.



(a) Baseline (b) Perceiver (c) Multiscale

Figure 4.3.: Case Study C: qualitative comparison of attention behavior across architecture variants.

Table 4.11.: External contextual baseline: fVLM Table 3 results on MedVL-CT69K (whole-report generation). Values are reported as in the source paper (0–100 scale).

Regime	Init	ACC	GREEN	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr
Frozen	IN	51.3	34.0	47.4	32.2	25.5	21.2	28.1	44.3	10.6
Frozen	SP	55.8	25.9	48.3	26.3	18.0	12.8	30.8	40.6	6.6
Frozen	MAE	50.7	21.6	49.0	27.1	18.6	13.1	30.5	41.6	6.1
Frozen	CLIP	57.3	33.4	49.7	28.8	20.7	15.5	31.0	42.2	9.6
Frozen	BIUD	58.4	33.7	47.1	30.4	23.4	18.9	29.1	44.2	13.9
Frozen	Merlin	58.8	34.2	49.5	31.4	23.9	19.0	30.0	43.8	14.3
Frozen	Ours	**61.5**	**37.2**	**50.7**	**32.2**	24.5	19.6	**31.3**	**45.1**	**14.9**
Fine-tuning	IN	58.0	33.4	49.4	29.8	21.9	16.9	30.4	43.6	10.0
Fine-tuning	SP	60.6	35.5	49.9	30.7	22.9	17.9	30.6	43.4	11.7
Fine-tuning	MAE	54.4	29.4	49.0	27.1	18.6	15.1	30.5	42.5	8.8
Fine-tuning	CLIP	62.0	37.6	49.9	31.9	24.3	19.5	30.7	44.7	14.3
Fine-tuning	BIUD	62.6	38.8	50.2	31.7	24.0	19.0	30.9	44.7	13.8
Fine-tuning	Merlin	63.0	39.2	50.7	33.3	25.8	20.9	31.1	46.0	**17.2**
Fine-tuning	Ours	**64.5**	**40.2**	**52.2**	**34.5**	**26.8**	**21.9**	**31.6**	**46.4**	17.1

the frozen regime, *Ours* is best on ACC, GREEN, BLEU-1, METEOR, ROUGE-L, CIDEr, and ties BLEU-2, while *IN* is best on BLEU-3 and BLEU-4.

For context, our best internal model (`medgemma_lora_vis_token_pos_embed`) corresponds to GREEN 34.8, BLEU-4 10.5, METEOR 34.4, ROUGE-L 37.9, and CIDEr 8.4 when scaled to 0–100. This is still not a direct model-to-model comparison: fVLM Table 3 uses MedVL-CT69K whole-report generation with a BERT-base decoder and paper-specific abnormality/diagnostic evaluation, while this thesis uses CT-RATE organ-decomposed generation with MedGemma and a unified RadEval export schema. A key strength of our evaluation remains the additional clinical structure metrics (RadGraph, RadCliQ, SRR-BERT), which are not reported in fVLM Table 3.

4.5. Chapter Summary and Minimal Limitations Note

The strongest internal model under current protocol is `medgemma_lora_vis_token_pos_embed`. The result pattern is consistent across RQ1–RQ3: interface stabilization is more beneficial than heavier architectural modules or training-policy complexity at the current compute budget.

The decoder axis (RQ4) remains a core comparison direction with finalized table structure and official artifacts. As of March 3, 2026, MedGemma metrics are complete; GPT-2/BioBART rows are pending the final standardized metric pass from official v3 generated reports.

Interpretation limits in this chapter are concise: uneven organ support in chest CT, partial metric-domain mismatch for organ-decomposed text, and non-equivalent external baseline protocol for fVLM. A full validity discussion is provided in Chapter 5.

5. Discussion

5.1. Discussion Scope

This chapter interprets the evidence from Chapter 4. The goal is to explain why the observed ranking emerged, which conclusions are robust, and where claims should be bounded.

5.2. Why Interface Design Dominated (Including Negative Ablations and Case-Study Evidence)

5.2.1. Empirical Signal

The strongest model, `medgemma_lora_vis_token_pos_embed`, improves core global metrics (GREEN, RadGraph, BLEU-4, ROUGE-L, BERTScore, RadCliQ) over the baseline. By contrast, heavier variants show mixed or negative deltas despite higher nominal capacity.

5.2.2. Mechanistic Interpretation

Three interface components appear decisive:

1. explicit visual token delimitation (sentinel wrapping),
2. consistent positional encoding over visual prefix tokens,
3. stabilized projection into decoder embedding scale.

Together they reduce ambiguity between visual context and textual instruction/response tokens.

5.2.3. Negative Ablations in Context

Perceiver-style resampling and multiscale fusion are not intrinsically weak. Under the current low effective batch, long context, and short retraining budget, their additional

complexity likely increases optimization burden more than it improves report-level factuality.

Curriculum and hard-example strategies also underperform the best interface variant. This indicates that weighting/sampling can refine learning but cannot fully compensate for an unstable vision-language interface.

5.2.4. Case-Study Support

Cross-attention overlays provide spatial evidence consistent with metric trends. When a model exhibits diffuse or anatomically implausible attention despite larger capacity, degraded factual metrics are easier to explain. This is why case studies are treated as supporting evidence for mechanism, not only as qualitative illustration.

5.2.5. Design Implication

For this task family, the practical order is: stabilize the visual-language interface first, verify organ grounding, then scale architectural complexity.

5.3. Metric Disagreement and Interpretation

Different metric families reward different behavior, so disagreement is expected:

- lexical metrics reward phrasing overlap,
- embedding metrics reward semantic proximity,
- clinical metrics reward factual and relational correctness.

CheXbert remains useful but is not treated as the sole clinical proxy for organ-decomposed CT text. Consistent with Section 4.2.1, model ranking prioritizes GREEN and RadGraph, then checks lexical-semantic coherence.

5.4. Generalization and Organ Coverage

The chest-CT setting yields strong thoracic support and sparse support for several upper-abdominal organs. This long-tail distribution affects both training dynamics and evaluation variance:

- sparse organs have noisier metrics,
- frequent thoracic patterns dominate global optimization,

- template bias risk increases for low-support organs.

Accordingly, organ-level conclusions are most reliable for high-support organs and should be interpreted cautiously for very sparse ones.

5.5. Decoder Family Axis

The decoder-comparison protocol is now structurally fixed (shared frontend, shared metric pipeline, shared artifact policy). MedGemma rows are complete; GPT-2/BioBART rows are pending only final standardized metric extraction from official v3 outputs. The remaining work is numerical completion rather than methodological redesign.

5.6. Relation to Prior Paradigms

Compared with many 2D report-generation settings, 3D CT introduces higher token pressure and stronger grounding demands. The results suggest that decoder scaling or heavier visual modules alone are insufficient if token semantics at the interface are not explicit and stable. This aligns with anatomy-guided trends in recent segmentation-aware and decomposition-aware literature.

5.7. Validity and Limits

5.7.1. Internal Validity

Internal validity is strongest for MedGemma-family comparisons because data interfaces and evaluation pipelines are controlled. Decoder-family claims remain provisional until GPT-2/BioBART final metric rows are inserted.

5.7.2. External Validity

Findings are specific to organ-decomposed chest-CT reporting and should not be over-generalized to other modalities or report styles without controlled replication.

5.7.3. Construct Validity

Construct validity is strengthened by metric bundles plus case studies, but still depends on imperfect automatic metrics. Human expert evaluation remains outside the current chapter scope.

5.8. Practical Implications and Synthesis

For practitioners, the evidence supports an interface-first recipe:

1. use explicit organ-conditioned supervision,
2. stabilize visual prefix semantics and projection scale,
3. run lightweight ablations before heavy architectural scaling,
4. use metric bundles plus qualitative spatial audits,
5. separate protocol-matched internal comparisons from external contextual baselines.

Overall synthesis: under realistic compute constraints, interface calibration is the dominant success factor in this organ-aware 3D CT report-generation setup.

6. Conclusion and Future Work

6.1. Conclusion

This thesis studied organ-aware radiology report generation from 3D chest CT with a unified vision-language pipeline. The core objective was to reduce the gap between rich volumetric information and decoder-side token constraints while preserving clinically meaningful report content.

The implemented framework combined organ-level supervision via report decomposition, organ-conditioned visual token extraction, and decoder adaptation under a consistent evaluation protocol. The most complete evidence currently comes from the MedGemma family, where multiple architectural and training ablations were compared systematically.

The main empirical conclusion is concise: in this setting, stable visual-language interfacing is more decisive than increasing architectural complexity. The strongest model variant used explicit visual-prefix structuring and delivered the most consistent improvements on the primary global metrics used in this thesis.

Overall, the thesis contributes:

- a reproducible end-to-end organ-aware 3D report generation setup,
- code-grounded ablation evidence on interface, architecture, and training strategy,
- a standardized evaluation workflow supporting multi-metric interpretation,
- a near-complete cross-decoder comparison framework (MedGemma/GPT-2/BioBART).

The work therefore advances both method (how to structure the task) and practice (how to implement and evaluate it reliably).

6.2. Take-Home Messages

The thesis can be reduced to four practical takeaways:

1. Organ-aware decomposition is not only a data-format choice; it is a training signal that makes failure analysis and controllable generation feasible.

2. In 3D CT report generation, token/interface design can matter more than adding architectural complexity.
3. Multi-metric evaluation is mandatory; single-metric optimization is too fragile for clinical text generation claims.
4. Reproducible artifact discipline (`generated_reports*.csv` + `metrics_final.csv`) is essential for trustworthy model comparison.

These messages are intended to be directly reusable by future work in volumetric medical vision-language systems, independent of decoder brand or training infrastructure.

6.3. Future Work

Future work is prioritized in three tracks.

6.3.1. Track 1: Complete the Current Thesis Axis

1. Finalize standardized metric extraction for official GPT-2/BioBART v3 outputs and populate the decoder comparison table.
2. Insert finalized cross-attention case-study figures to complete the quantitative + qualitative evidence pair.
3. Add compact significance reporting (e.g., bootstrap confidence intervals) for key global metrics.

This track is short-horizon and can be completed without redesigning the model stack. Its purpose is to close the evidence loop of the current thesis version.

6.3.2. Track 2: Improve Model Robustness and Efficiency

1. Run controlled token-budget sweeps (1/2/4/8 tokens per organ) under matched compute.
2. Compare mask downsampling strategies (area interpolation vs max-pooling) for mask-to-token alignment.
3. Ablate bridge stabilization choices (with/without variance matching and visual positional embeddings).

4. Compare loss composition (pure LM vs LM+alignment) on branches where alignment is implemented.
5. Compare sampling policies (uniform vs organ-weighted vs hard-example weighted).
6. Explore organ-adaptive token allocation and uncertainty-aware handling for sparse organs.

This track targets the main technical bottlenecks identified by the ablations: token efficiency, interface stability, and long-tail organ robustness.

6.3.3. Track 3: Strengthen Clinical Validity

1. Add human evaluation with radiologist-centered error taxonomy.
2. Expand evaluation to temporal/longitudinal scenarios and clinically realistic reading workflows.
3. Introduce safety-oriented post-generation checks for contradiction and hallucination control.

This track is required for translational credibility. It shifts the system from benchmark-oriented progress to clinically meaningful utility and safety assessment.

6.4. Final Remark

The thesis shows that organ-aware 3D report generation is tractable when anatomical structure is treated as a first-class modeling constraint. With the current framework and results, the project is technically mature enough to complete cross-decoder benchmarking and move toward stronger clinical validation.

A. Metric Equations

This appendix provides the formal definitions of lexical metrics referenced in Chapter 4.

A.1. BLEU

For candidate report c and references $\{r\}$:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right), \tag{A.1}$$

where p_n is modified n-gram precision and w_n are n-gram weights. The brevity penalty is:

$$\text{BP} = \begin{cases} 1 & c_{len} > r^*_{len} \\ \exp\left(1 - \frac{r^*_{len}}{c_{len}}\right) & \text{otherwise.} \end{cases} \tag{A.2}$$

A.2. ROUGE-L

Using longest common subsequence (LCS):

$$R_{LCS} = \frac{\text{LCS}(c,r)}{|r|}, \qquad P_{LCS} = \frac{\text{LCS}(c,r)}{|c|},$$

$$F_{LCS} = \frac{(1+\beta^2) R_{LCS} P_{LCS}}{R_{LCS} + \beta^2 P_{LCS}}. \tag{A.3}$$

A.3. METEOR

$$F_{\text{mean}} = \frac{(1+\alpha)PR}{R + \alpha P}, \tag{A.4}$$

$$\text{METEOR} = F_{\text{mean}} \cdot (1 - \text{Penalty}). \tag{A.5}$$

A.4. CIDEr

With TF-IDF-weighted n-gram vectors:

$$\mathrm{CIDEr}_n(c, r) = \frac{1}{M} \sum_{i=1}^{M} \frac{\mathbf{g}^n(c) \cdot \mathbf{g}^n(r_i)}{\|\mathbf{g}^n(c)\| \|\mathbf{g}^n(r_i)\|}. \tag{A.6}$$

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